

CS5239 – Performance Analysis of Computer Systems

Admin Information

Sem 1 AY 2026/27

Teaching Staff – Senior Lecturer

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- Comments / suggestions welcome
- Email in advance to book a consultation slot

Overview

- CS5239 teaches the challenging field of systems performance
- Systems performance studies the performance of an entire computer system
 - All major software and hardware components
 - Anything in the data path, from storage devices to application software
 - For distributed systems, multiple servers and applications
- Typical goals of systems performance
 - Improve the end-user experience by **reducing latency**
 - Reduce computing cost by eliminating inefficiencies, improving system throughput, and general tuning

Performance analysis is necessary

- Our applications use
 - Hardware components – several cores, memories, caches, network cards, etc
 - Operating system(s)
 - Compilers
 - Programming languages
 - Interconnection networks (internet, proprietary, etc.)
- Every entity has an impact on the performance observed when running the applications

Not that simple...

- Practice performance analysis while you write your software
 - Measurement
 - Analytical modeling
 - Simulation
- You should have an understanding about your hardware system, network, operating system, compilers (if any) needed when running your software

Performance analysis is part of software developers' lives nowadays.

Systems performance activities in the industry

1. Setting performance objectives and performance modeling for a future product
2. Performance characterization of prototype software and hardware
3. Performance analysis of in-development products in a test environment
4. Non-regression testing for new product versions
5. Benchmarking product releases
6. Proof-of-concept testing in the target production environment
7. Performance tuning in production
8. Monitoring of running production software
9. Performance analysis of production issues
10. Incident reviews for production issues
11. Performance tool development to enhance production analysis

Learning Outcomes

By the end of the module, the students should be able to

- Explain and interpret key performance metrics, including latency, utilization, and saturation.
- Analyse and evaluate system performance using measurement techniques and analytical models to draw evidence-based conclusions.
- Interpret and model system performance using principles from queueing theory and operational analysis.
- Evaluate and justify the use of different performance analysis methodologies (e.g., USE method, workload characterization, latency analysis, static tuning, bottleneck analysis) for specific system scenarios.
- Optimize system performance by identifying bottlenecks and designing appropriate performance improvements.

Why study this module?

- Systems performance is done by a variety of job roles
 - System administrators, site reliability engineers, application developers, network engineers, database administrators, web administrators, and other support staff
 - Sometimes, finding the root cause or contributing factors requires a cooperative effort from more than one team within a company
- Performance engineers – primary activity is performance
 - Holistic study of the environment
 - Find and develop better tooling for performance analysis and capacity planning across the whole environment

Assessment – 100% CA

- Individual:
 - 20% - Test 1 (Tue, week 6, at the lecture, in-person, open book)
 - 30% - Test 2 (Tue, week 12, at the lecture, in-person, open book)
- Team of 5:
 - 10% - Paper review (due on Tue, week 10)
 - 5% - Presentation of the project or paper (Tue, week 13)
 - 35% - Project (due on Tue, reading week)

Tentative schedule – aka Topics covered

Week	Lecture (Tue, 18:30pm) – LT16	Deadlines
1	L0: Course Admin L1: Introduction	
2	L2: Methodologies and Capacity Planning Lstats (only lecture notes): Review of Probability and Statistics Concepts	
3	L3: Introduction to Queueing Theory	
4	L4: Markovian Queueing Models	
5	L5: Operational Analysis	
6	L6: Background in OS Test 1 (open-book, in-person)	Tue, 18:30 - Test 1
Recess		
7	L7: Observability Tools	
8	L8: Resources Analysis (System)	
9	L9: Monitoring and Visualization	
10	L10: Performance analysis on GPGPU	
11	L11: Applications and Benchmarking L12: Case studies	Tue, 18:30 - Paper review due
12	Test 2 (open-book, in-person)	Tue, 18:30 - Test 2
13	Student Presentations	Tue, 18:30 - Presentations
Reading	No exam CS5239 L0 - Admin	Tue, 18:30 - Project due

References

[Jain] “The Art of Computer Systems Performance Analysis: Techniques for Experimental Design, Measurement, Simulation, and Modeling” by Raj Jain, 1991.

[Gregg] “Systems Performance: Enterprise and the Cloud” by Brendan Gregg 2nd Edition, 2020.

Pre-requisites

- What do we assume you know
 - Operating systems (implies basic knowledge of computer architecture)
 - Basic statistics concepts
 - Basic probability concepts
 - Use a shell console (intermediate level)
 - Use a systems programming language (C/C++/Rust, Python might work as well) (intermediate level)
- Ideally, you should install natively a Linux distribution (Ubuntu) on your computer, or get (remote) access to such a computer
 - Important in the second part of the semester

Disclaimer

- Send me your comments anytime
 - Be constructive in your comments
 - Anonymous survey on Canvas always open and ready for your inputs
- Targeting to publish:
 - Lectures on Monday morning
 - Practice Questions for test 1 in week 4
 - Papers for review in week 7
 - Project topics in week 7
- Delays in publishing materials might happen
 - We did not forget about it, but most likely it is not ready yet